

An exact coefficient formula for Ramanujan's theta function

A self-contained proof of both formulas in OEIS A260306

Research note

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Abstract

For positive integers N , define $\theta(N)$ by $\sum_{k=0}^{N-1} N^k/k! + \theta(N)N^N/N! = e^N/2$. The numerators of the coefficients in its asymptotic expansion form OEIS A260306. We prove both explicit formulas recorded in that entry: a double sum involving Stirling numbers of the second kind and an equivalent triple sum involving only binomial coefficients, factorials, and integer powers. The proof first converts the defining expression into the difference of two Laplace integrals. An analytic change of variable and a direct remainder estimate establish the expansion to every order. Lagrange inversion then reduces each coefficient to a single coefficient extraction. A finite change-of-basis identity turns that extraction into the exact OEIS sums, including all their signs and factorial factors. We also derive the recurrence used in the entry, a one-sum formula using associated Stirling numbers, and reproducible exact-arithmetic checks. The argument is a proof for every nonnegative coefficient index, not an extrapolation from computed initial values.

1 The statement and its normalization

For a positive integer N , let

$$\sum_{k=0}^{N-1} \frac{N^k}{k!} + \theta(N) \frac{N^N}{N!} = \frac{e^N}{2}. \quad (1.1)$$

We use N for the large argument and n for a coefficient index. Thus the expansion under discussion is

$$\theta(N) \sim \sum_{n \geq 0} \frac{c_n}{N^n}. \quad (1.2)$$

In this article, (1.2) means that, for each fixed integer $R \geq 1$,

$$\theta(N) = \sum_{n=0}^{R-1} \frac{c_n}{N^n} + O_R(N^{-R}) \quad (N \longrightarrow +\infty). \quad (1.3)$$

It is not being used as an assertion that the infinite series converges.

Write $c_n = a(n)/b(n)$ in lowest terms, with $b(n) > 0$. The OEIS entries A260306 and A065973 specify $a(n)$ and $b(n)$, respectively [1, 2]. Their initial coefficients are

$$c_0 = \frac{1}{3}, \quad c_1 = \frac{4}{135}, \quad c_2 = -\frac{8}{2835}, \quad c_3 = -\frac{16}{8505}, \quad c_4 = \frac{8992}{12629925}. \quad (1.4)$$

The formulas in the entry are formulas for the *rational coefficients* c_n , not for the integer numerators before reduction.

Let $S(L, j)$ denote a Stirling number of the second kind: the number of partitions of an L -element set into j nonempty blocks. In particular, $S(0, 0) = 1$ and $S(L, 0) = 0$ for $L > 0$.

Theorem 1.1 (The two OEIS formulas). *The expansion (1.3) exists to every order. For every integer $n \geq 0$, its coefficient c_n satisfies*

$$c_n = 2^n (3n+2)! \sum_{i=1}^{2n+1} \sum_{j=1}^i \frac{(-1)^{j+1} 2^i S(2n+i+j+1, j)}{(2n+i+j+1)! (2n-i+1)! (i-j)! (n+i+1)!}. \quad (1.5)$$

Equivalently,

$$c_n = \frac{2^n (3n+2)!}{(2n+1)!} \sum_{i=1}^{2n+1} \sum_{j=1}^i \sum_{k=1}^j \frac{(-1)^{k+1} 2^i k^{2n+i+j+1} \binom{2n+1}{i} \binom{i}{j} \binom{j}{k}}{(2n+i+j+1)! (n+i+1)!}. \quad (1.6)$$

Consequently, the reduced numerators of either right-hand side are A260306, and the reduced positive denominators are A065973.

The finite sums (1.5)–(1.6) are the formulas displayed in A260306 [1]. O'Sullivan gives a broader study of Ramanujan's approximation using saddle-point methods, including other closed forms for its coefficients [3]. Our purpose here is to give a direct, self-contained derivation of these *particular* OEIS expressions. No claim of novelty for Ramanujan's expansion or its general coefficient theory is needed.

The decisive intermediate identity will be

$$c_n = -2^n n! [z^{2n+1}] \left(\frac{z^2}{2(e^z - 1 - z)} \right)^{n+1}. \quad (1.7)$$

The quotient in parentheses has a removable singularity at zero and constant term 1. Sections 2–4 establish (1.7) analytically; Sections 5–7 convert it into the advertised sums by finite algebra.

2 An exact integral representation

For $x > 0$ and $u > 0$, define

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt, \quad \Gamma(x, u) = \int_u^\infty t^{x-1} e^{-t} dt.$$

Integration by parts, starting with $\Gamma(1, u) = e^{-u}$, gives

$$\Gamma(N, u) = (N-1)! e^{-u} \sum_{k=0}^{N-1} \frac{u^k}{k!} \quad (N \in \mathbb{Z}_{\geq 1}).$$

Setting $u = N$ in this identity and rearranging (1.1) yields

$$\theta(N) = \frac{N e^N}{N^N} \left(\frac{1}{2} \Gamma(N) - \Gamma(N, N) \right). \quad (2.1)$$

The right-hand side also defines a natural extension to every real $N > 0$. We use that extension in the analysis below; its restriction to positive integers is exactly the original function.

In both gamma integrals substitute $t = Ne^x$. Since

$$t^{N-1} e^{-t} dt = N^N e^{-N} \exp\{-N(e^x - 1 - x)\} dx,$$

formula (2.1) becomes

$$\theta(N) = \frac{N}{2} \left(\int_{-\infty}^0 e^{-NF(x)} dx - \int_0^{\infty} e^{-NF(x)} dx \right), \quad F(x) = e^x - 1 - x. \quad (2.2)$$

This equality is exact. The common exponential scale and all gamma factors have disappeared. In particular, no separate use of Stirling's expansion for $N!$ is required.

The function F is positive away from zero, with

$$F(0) = F'(0) = 0, \quad F''(0) = 1, \quad F'(x) = e^x - 1.$$

Thus each integral is concentrated near the same nondegenerate minimum. Their difference will cancel the terms that would otherwise produce half-integer powers of N^{-1} .

3 The inverse change of variable and the remainder

Define the entire function with a removable value at zero

$$A(z) = \frac{2(e^z - 1 - z)}{z^2} = \sum_{r=0}^{\infty} \frac{2z^r}{(r+2)!} = 1 + \frac{z}{3} + \frac{z^2}{12} + \frac{z^3}{60} + \cdots. \quad (3.1)$$

Near $z = 0$, choose the analytic square root of $A(z)$ with value 1. The function $w = z\sqrt{A(z)}$ then has derivative 1 at zero, so it has a unique local analytic inverse

$$z = \psi(w) = \sum_{m=1}^{\infty} b_m w^m, \quad b_1 = 1, \quad F(\psi(w)) = \frac{w^2}{2}. \quad (3.2)$$

On the real line the same change of variable can be written globally as

$$w = \operatorname{sgn}(x) \sqrt{2F(x)}.$$

It is strictly increasing: away from zero its derivative is $F'(x)/w > 0$, and at zero its derivative is 1. Its limits at the two ends of the real line are $-\infty$ and $+\infty$, respectively. It therefore has a global real inverse, agreeing with the analytic inverse ψ near zero. Differentiating (3.2) gives

$$\psi'(w) = \frac{w}{e^{\psi(w)} - 1} \quad (w \neq 0). \quad (3.3)$$

Substitution into (2.2), followed by reflection of the negative half-axis, gives the exact identity

$$\theta(N) = \frac{N}{2} \int_0^{\infty} e^{-Nw^2/2} (\psi'(-w) - \psi'(w)) dw. \quad (3.4)$$

Proposition 3.1 (All-orders expansion). *For every fixed integer $R \geq 1$, (1.3) holds with*

$$c_n = -2^{n+1} (n+1)! b_{2n+2}, \quad n \geq 0. \quad (3.5)$$

Proof. Choose $\delta > 0$ small enough that the power series for ψ is analytic on a neighborhood of $[-\delta, \delta]$. Its derivative difference has the odd Taylor expansion

$$\psi'(-w) - \psi'(w) = -2 \sum_{n=0}^{R-1} (2n+2)b_{2n+2}w^{2n+1} + O_R(w^{2R+1}) \quad (0 \leq w \leq \delta). \quad (3.6)$$

The required Gaussian moments are elementary:

$$\int_0^\infty e^{-Nw^2/2} w^{2n+1} dw = \frac{2^n n!}{N^{n+1}}. \quad (3.7)$$

Indeed, substitute $v = Nw^2/2$.

We now justify replacing the local expansion by these moments. For $w \geq \delta$, the negative inverse branch satisfies

$$0 < \psi'(-w) = \frac{w}{1 - e^{\psi(-w)}} \leq \frac{w}{1 - e^{\psi(-\delta)}}.$$

For the positive branch, $e^{\psi(w)} - 1 = \psi(w) + w^2/2 \geq w^2/2$, so

$$0 < \psi'(w) \leq \frac{2}{w} \leq \frac{2w}{\delta^2}.$$

Consequently,

$$\frac{N}{2} \int_\delta^\infty e^{-Nw^2/2} |\psi'(-w) - \psi'(w)| dw \leq C_\delta N \int_\delta^\infty w e^{-Nw^2/2} dw = C_\delta e^{-N\delta^2/2}.$$

Thus the nonlocal part is exponentially small.

On $[0, \delta]$, the remainder in (3.6) contributes at most

$$C_R N \int_0^\delta e^{-Nw^2/2} w^{2R+1} dw \leq C_R 2^R R! N^{-R}.$$

For each of the finitely many polynomial terms, extending its integral from δ to infinity changes the answer by an exponentially small quantity. One can see this directly by substituting $v = w^2/2$ and integrating $e^{-Nv} v^n$ by parts a finite number of times.

Finally, inserting (3.6) and (3.7) into (3.4) gives

$$\frac{N}{2} \cdot (-2)(2n+2)b_{2n+2} \frac{2^n n!}{N^{n+1}} = \frac{-2^{n+1}(n+1)!b_{2n+2}}{N^n}.$$

The stated coefficients and remainder follow. □

The parity cancellation in (3.6) explains why only integer powers N^{-n} occur. The argument proves the expansion for real $N \rightarrow +\infty$, hence in particular along the positive integers. Its coefficients are unique: if two expansions first differ at index n , subtracting them and multiplying by N^n gives a nonzero constant that would have to tend to zero.

4 Lagrange inversion and the coefficient identity

We include the particular form of Lagrange inversion that is needed, so that no coefficient-extraction step is left implicit.

Lemma 4.1 (Inverse-series coefficient formula). *Suppose $w = z\phi(z)$, where ϕ is analytic near zero and $\phi(0) = 1$. Let $z = g(w)$ be its local inverse. For every $m \geq 1$,*

$$[w^m]g(w) = \frac{1}{m}[z^{m-1}]\phi(z)^{-m}. \quad (4.1)$$

Proof. Differentiation and a change of variable in the residue give

$$\begin{aligned} m[w^m]g(w) &= \operatorname{Res}_{w=0} \frac{g'(w)}{w^m} dw \\ &= \operatorname{Res}_{z=0} \frac{1}{(z\phi(z))^m} dz = [z^{m-1}]\phi(z)^{-m}. \end{aligned}$$

All residues can be taken on sufficiently small contours on which the analytic change of variable is invertible. $\square$

Apply the lemma with $\phi(z) = \sqrt{A(z)}$ and $m = 2n + 2$. Then

$$b_{2n+2} = \frac{1}{2n+2}[z^{2n+1}]A(z)^{-n-1}.$$

Together with (3.5), this proves the key identity (1.7). In particular, every c_n is rational, since the power series of A has rational coefficients.

Corollary 4.2 (Residue form). *For every $n \geq 0$,*

$$c_n = -\frac{n!}{2} \operatorname{Res}_{z=0} \frac{dz}{(e^z - 1 - z)^{n+1}}. \quad (4.2)$$

Proof. In (1.7), replace $A(z)^{-n-1}$ by $z^{2n+2}/(2^{n+1}F(z)^{n+1})$. The coefficient of z^{2n+1} is 2^{-n-1} times the coefficient of z^{-1} in $F(z)^{-n-1}$. $\square$

Formula (1.7) is already a compact exact answer. The remaining work is to show why its finite expansion has precisely the less obvious shape recorded in OEIS.

5 A finite change of basis for negative powers

The factorial $(3n+2)!$ and the factor $n+i+1$ in the OEIS formulas come from a general finite algebraic identity.

Lemma 5.1 (Finite negative-power identity). *Let $p \geq 1$ and $M \geq 1$ be integers, and let $B(z)$ be a formal power series over $\mathbb{Q}$ or $\mathbb{C}$ with $B(0) = 1$. Then*

$$[z^M]B(z)^{-p} = \frac{(p+M)!}{(p-1)!} \sum_{i=1}^M \frac{(-1)^i [z^M]B(z)^i}{i! (M-i)! (p+i)}. \quad (5.1)$$

Proof. Since $B(z) - 1$ has zero constant term, only powers at most M can affect a coefficient of degree M . Therefore

$$B(z)^{-p} \equiv \sum_{r=0}^M (-1)^r \binom{p+r-1}{r} (B(z) - 1)^r \pmod{z^{M+1}}. \quad (5.2)$$

Expand $(B - 1)^r$ in powers of B and interchange the two finite sums. The coefficient multiplying B^i is

$$\begin{aligned} \sum_{r=i}^M (-1)^r \binom{p+r-1}{r} \binom{r}{i} (-1)^{r-i} &= (-1)^i \sum_{r=i}^M \binom{p+r-1}{r} \binom{r}{i} \\ &= (-1)^i \binom{p+i-1}{i} \sum_{r=i}^M \binom{p+r-1}{r-i} \\ &= (-1)^i \binom{p+i-1}{i} \binom{p+M}{M-i} \\ &= \frac{(-1)^i (p+M)!}{(p-1)! i! (M-i)! (p+i)!}. \end{aligned}$$

The penultimate equality is the hockey-stick identity, obtained by repeated application of Pascal's identity. The term $i = 0$ multiplies $B^0 = 1$ and has zero coefficient of z^M , because $M \geq 1$. Taking $[z^M]$ proves the lemma. $\square$

Remark 5.2 (An integral interpretation of the denominator). The polynomial in X produced by the proof can also be written as

$$T_{p,M}(X) = \frac{(p+M)!}{(p-1)!M!} \int_0^1 t^{p-1} (1-Xt)^M dt. \quad (5.3)$$

Expanding $(1-Xt)^M$ shows that the coefficient of X^i includes $\int_0^1 t^{p+i-1} dt = 1/(p+i)$. Thus the denominator in (5.1) has a simple finite-polynomial origin.

Now put

$$M = 2n + 1, \quad p = n + 1, \quad B(z) = A(z). \quad (5.4)$$

Multiplying (5.1) by $-2^n n!$, as prescribed by (1.7), gives

$$c_n = 2^n (3n+2)! \sum_{i=1}^M \frac{(-1)^{i+1} [z^M] A(z)^i}{i! (M-i)! (n+i+1)!}. \quad (5.5)$$

Here $(p+M)! = (3n+2)!$ and $(p-1)! = n!$, which cancels the $n!$ from (1.7). These are exactly the factorials needed in the target.

6 Extraction with Stirling numbers: the double sum

The exponential generating function for Stirling numbers is

$$(e^z - 1)^j = j! \sum_{L \geq 0} S(L, j) \frac{z^L}{L!}. \quad (6.1)$$

For completeness, expanding $(e^z - 1)^j$ gives, for $L, j \geq 1$,

$$j! S(L, j) = \sum_{k=0}^j (-1)^{j-k} \binom{j}{k} k^L. \quad (6.2)$$

The right-hand side counts surjections from an L -element set to a j -element set by inclusion-exclusion; dividing by $j!$ forgets the labels of the nonempty fibers. This proves (6.1) and (6.2) directly.

From (3.1), for $i \geq 1$,

$$\begin{aligned}
 [z^M]A(z)^i &= 2^i [z^{M+2i}](e^z - 1 - z)^i \\
 &= 2^i \sum_{j=0}^i (-1)^{i-j} \binom{i}{j} [z^{M+i+j}](e^z - 1)^j \\
 &= 2^i \sum_{j=1}^i (-1)^{i-j} \binom{i}{j} \frac{j! S(M+i+j, j)}{(M+i+j)!}.
 \end{aligned} \tag{6.3}$$

The second line follows by writing $e^z - 1 - z = (e^z - 1) - z$. In the third line, the $j = 0$ term vanishes because it asks for a positive coefficient of the constant series 1.

Insert (6.3) into (5.5). Two elementary simplifications are responsible for the final form:

$$(-1)^{i+1}(-1)^{i-j} = (-1)^{j+1}, \quad \frac{1}{i!} \binom{i}{j} j! = \frac{1}{(i-j)!}.$$

It follows that

$$c_n = 2^n (3n+2)! \sum_{i=1}^M \sum_{j=1}^i \frac{(-1)^{j+1} 2^i S(M+i+j, j)}{(M+i+j)! (M-i)! (i-j)! (n+i+1)!}. \tag{6.4}$$

Substituting $M = 2n + 1$ is exactly (1.5). This proves the Stirling-number formula for every $n \geq 0$.

All bounds and factorials are valid at the endpoints: $M \geq 1$, $1 \leq j \leq i \leq M$, and therefore $M - i$ and $i - j$ are nonnegative. No zero factorial is omitted, and $n + i + 1$ is strictly positive.

7 From the double sum to the triple sum

Let $L = M + i + j$. This exponent is positive, so the $k = 0$ term in (6.2) is zero. Multiplying that identity by $(-1)^{j+1}$ gives

$$(-1)^{j+1} j! S(L, j) = \sum_{k=1}^j (-1)^{k+1} \binom{j}{k} k^L. \tag{7.1}$$

Indeed, $(-1)^{j+1}(-1)^{j-k} = (-1)^{k+1}$.

Also,

$$\frac{S(L, j)}{(M-i)! (i-j)!} = \frac{1}{M!} \binom{M}{i} \binom{i}{j} j! S(L, j). \tag{7.2}$$

Substituting (7.2) and (7.1) into (6.4) yields

$$\begin{aligned}
 c_n &= \frac{2^n (3n+2)!}{M!} \sum_{i=1}^M \sum_{j=1}^i \sum_{k=1}^j \\
 &\quad \times \frac{(-1)^{k+1} 2^i k^{M+i+j} \binom{M}{i} \binom{i}{j} \binom{j}{k}}{(M+i+j)! (n+i+1)!}.
 \end{aligned} \tag{7.3}$$

Replacing M by $2n + 1$ proves (1.6) and completes the proof of Theorem 1.1. Since every exponent L here is positive, no convention concerning 0^0 is involved.

The roles of the three summation indices are now explicit. The index i arises from replacing a negative power of A by a finite combination of positive powers. The index j comes from expanding $((e^z - 1) - z)^i$. The index k is the inclusion–exclusion index that eliminates the Stirling number. In particular, equivalence of the double and triple sums alone would not identify them with Ramanujan's coefficients; the integral and inversion arguments supply that essential identification.

8 The recurrence in the OEIS entry

A recurrence provides a compact way to compute a long initial segment. It is also useful to explain the separate treatment of the constant term in the OEIS program [1].

Define

$$u(w) = e^{\psi(w)} - 1. \quad (8.1)$$

The inverse equation $e^{\psi(w)} - 1 - \psi(w) = w^2/2$ implies the especially simple relation

$$u(w) = \psi(w) + \frac{w^2}{2}. \quad (8.2)$$

Equivalently, $u - \log(1 + u) = w^2/2$ near zero. Let

$$H(w) = u'(w) = \sum_{m \geq 0} h_m w^m, \quad h_0 = 1. \quad (8.3)$$

Differentiating the latter inverse equation gives

$$u(w)H(w) = w(1 + u(w)). \quad (8.4)$$

Since $u(w) = \sum_{j \geq 0} h_j w^{j+1}/(j+1)$, comparison of coefficients of w^{m+1} gives, for $m \geq 1$,

$$\sum_{j=0}^m \frac{h_j h_{m-j}}{j+1} = \frac{h_{m-1}}{m}.$$

Separating the terms $j = 0$ and $j = m$ yields

$$h_m = \frac{\frac{h_{m-1}}{m} - \sum_{j=1}^{m-1} \frac{h_{m-j} h_j}{j+1}}{1 + \frac{1}{m+1}}, \quad h_0 = 1. \quad (8.5)$$

Its beginning is

$$H(w) = 1 + \frac{2}{3}w + \frac{1}{12}w^2 - \frac{2}{135}w^3 + \frac{1}{864}w^4 + \frac{1}{2835}w^5 + \cdots.$$

By (8.2), $H(w) = \psi'(w) + w$, hence

$$h_{2n+1} = (2n+2)b_{2n+2} + \delta_{n,0}.$$

Formula (3.5) is therefore equivalent to

$$c_n = \delta_{n,0} - 2^n n! h_{2n+1}. \quad (8.6)$$

For $n = 0$, this says $c_0 = 1 - h_1 = 1/3$. For $n \geq 1$, it says $c_n = -(2n)!! h_{2n+1}$, since $(2n)!! = 2^n n!$. These are precisely the recurrence and the constant-term exception underlying the Maple code in A260306. Forgetting the Kronecker delta in (8.6) would give $-2/3$ instead of $1/3$ at index zero.

8.1 A second coefficient algorithm

There is a separate exact computation directly from (1.7). For fixed $p \geq 1$, write

$$A(z)^{-p} = \sum_{m \geq 0} f_m z^m, \quad f_0 = 1, \quad a_k = [z^k]A(z) = \frac{2}{(k+2)!}.$$

The differential identity $A(A^{-p})' = -pA'A^{-p}$ gives

$$f_m = -\frac{1}{m} \sum_{k=1}^m (m + (p-1)k) a_k f_{m-k} \quad (m \geq 1). \quad (8.7)$$

Use $p = n+1$ and $m = 2n+1$, then multiply by $-2^n n!$. This algorithm is independently implemented in the verification code, alongside (8.5), both OEIS sums, and the next corollary.

To obtain all coefficients through c_K , recurrence (8.5) uses $O(K^2)$ rational arithmetic operations. A single double-sum coefficient has $O(n^2)$ summands after its Stirling table is available; the literal triple sum has $O(n^3)$ summands. These are arithmetic-operation counts, not bit-complexity bounds. Numerators and denominators grow, and alternating sums can involve severe cancellation, so the exact algorithms use integers and rational numbers rather than floating-point arithmetic.

9 A one-sum corollary with associated Stirling numbers

Let $\left\{ \begin{smallmatrix} L \\ k \end{smallmatrix} \right\}_{\geq 3}$ count partitions of an L -element set into k blocks, each of size at least three. The associated exponential generating function is

$$\frac{(e^z - 1 - z - z^2/2)^k}{k!} = \sum_{L \geq 0} \left\{ \begin{smallmatrix} L \\ k \end{smallmatrix} \right\}_{\geq 3} \frac{z^L}{L!}. \quad (9.1)$$

This follows by forming k labeled blocks of sizes at least three and then dividing by $k!$ to remove their order. O'Sullivan's treatment also relates the asymptotic coefficients to associated Stirling numbers [3]; here the needed identity follows immediately from (1.7).

Corollary 9.1. *For every $n \geq 0$,*

$$c_n = \sum_{k=1}^{2n+1} \frac{(-1)^{k+1} 2^{n+k} (n+k)!}{(2n+2k+1)!} \left\{ \begin{smallmatrix} 2n+2k+1 \\ k \end{smallmatrix} \right\}_{\geq 3}. \quad (9.2)$$

Proof. Put $M = 2n+1$. Instead of changing basis as in Lemma 5.1, expand A^{-n-1} directly in powers of $A-1$:

$$[z^M]A(z)^{-n-1} = \sum_{k=1}^M (-1)^k \binom{n+k}{k} [z^M](A(z)-1)^k.$$

Since $A(z)-1 = 2(e^z - 1 - z - z^2/2)/z^2$, (9.1) gives

$$[z^M](A(z)-1)^k = \frac{2^k k!}{(M+2k)!} \left\{ \begin{smallmatrix} M+2k \\ k \end{smallmatrix} \right\}_{\geq 3}.$$

Multiplication by $-2^n n!$ proves the result. The upper bound $k \leq M$ also follows combinatorially: $M+2k$ elements cannot form k blocks of size at least three unless $M+2k \geq 3k$. $\square$

For exact computation, these associated numbers satisfy, for $L, k \geq 1$,

$$\left\{ \begin{matrix} L \\ k \end{matrix} \right\}_{\geq 3} = k \left\{ \begin{matrix} L-1 \\ k \end{matrix} \right\}_{\geq 3} + \binom{L-1}{2} \left\{ \begin{matrix} L-3 \\ k-1 \end{matrix} \right\}_{\geq 3}. \quad (9.3)$$

Use the usual zero conventions outside the allowed range and $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\}_{\geq 3} = 1$. To prove the recurrence, distinguish one element. Either its block has at least four elements, in which case removing it leaves one of k eligible blocks, or its block has exactly three elements, in which case choose its two companions and partition the remaining $L-3$ elements. This supplies another integer-based check of (9.2).

10 Initial coefficients and worked cases

The first case is visible without any summation. From (3.1),

$$[z]A(z)^{-1} = -\frac{1}{3}, \quad c_0 = -[z]A(z)^{-1} = \frac{1}{3}.$$

The double sum also has only one term when $n = 0$: $i = j = 1$, giving

$$2! \frac{2S(3, 1)}{3!0!0!2} = \frac{1}{3}.$$

Thus the displayed OEIS sum itself needs no exceptional rule at $n = 0$.

For $n = 1$, put $a_1 = 1/3$, $a_2 = 1/12$, and $a_3 = 1/60$. The binomial expansion of A^{-2} yields

$$\begin{aligned} [z^3]A(z)^{-2} &= -2a_3 + 6a_1a_2 - 4a_1^3 \\ &= -\frac{1}{30} + \frac{1}{6} - \frac{4}{27} = -\frac{2}{135}. \end{aligned}$$

Therefore $c_1 = -2[z^3]A(z)^{-2} = 4/135$, confirming both the factor and the sign in the general formula.

n	Reduced numerator $a(n)$	Positive denominator $b(n)$
0	1	3
1	4	135
2	-8	2835
3	-16	8505
4	8992	12629925
5	334144	492567075
6	-698752	1477701225
7	-23349012224	39565450299375
8	1357305243136	2255230667064375
9	6319924923392	6765692001193125
10	-8773495082018816	7002491221234884375

Table 1: Coefficients generated by exact arithmetic from the proved recurrence. The displayed values agree with the corresponding prefixes of A260306 and A065973 [1, 2].

The table illustrates why it is important to reduce each rational coefficient before extracting its numerator or denominator. Neither the individual summands nor the unreduced factorial prefactor alone determines the two OEIS sequences.

11 Reproducible computation and a numerical illustration

The accompanying Python code uses the standard-library `fractions.Fraction` class for every exact coefficient computation. The executed checks are recorded in `artifacts/checks.json`. Specifically, the inverse-function recurrence generated c_0 through c_{117} . For $0 \leq n \leq 20$, these coefficients were compared exactly with the inverse-power algorithm, the double sum, and the associated Stirling-number sum. For $0 \leq n \leq 10$, they were also compared with the literal triple sum. Every comparison passed.

An external-data check compared the first 17 computed numerators and the first 14 computed denominators with prefixes transcribed from the consulted OEIS entries. The precise prefixes, source addresses, and access date are in `artifacts/reference_prefix.json`. All of those comparisons passed as well. The 118 generated coefficients should not be confused with 118 independently checked OEIS values: the external comparison in this run covers only the stated prefixes. The optional live b-file comparison is available in the script but was not completed in the restricted execution environment.

To rerun the exact checks from the archive's root directory, use

```
python3 code/verify.py
```

No third-party package and no network access is needed for that command. The output includes `coefficients.csv`, containing separately the index, reduced numerator, and reduced positive denominator. These checks are finite consistency tests; the all-index assertion is established by Theorem 1.1.

11.1 Checking the asymptotic scale

A separate script evaluates $\theta(N)$ in two ways: from the scaled finite-sum definition and from the incomplete-gamma expression (2.1). In the executed run, `mpmath` version 1.3.0 was used at 100 decimal digits of working precision. The two evaluations differed by less than 7.18×10^{-99} for all tested $N \in \{5, 10, 20, 50, 100, 200, 500, 1000\}$.

Let

$$P_4(N) = \frac{1}{3} + \frac{4}{135N} - \frac{8}{2835N^2} - \frac{16}{8505N^3}.$$

Applying (1.3) with $R = 6$ gives

$$N^4(\theta(N) - P_4(N)) = c_4 + \frac{c_5}{N} + O(N^{-2}), \quad (11.1)$$

where

$$c_4 = \frac{8992}{12629925} \approx 0.000711959888914621.$$

Table 2 illustrates the limit in (11.1). The numbers are high-precision computations rounded for display, not interval-certified error bounds; the analytic proof above supplies the rigorous asymptotic assertion.

The optional numerical run is reproduced with

```
python3 -m pip install -r requirements-optional.txt
python3 code/numerical_check.py
```

The README describes the remaining files and the PDF build command.

N	$N^4(\theta(N) - P_4(N))$	Ratio to c_4
10	0.000774546454009	1.087907431400
20	0.000744626622858	1.045882829148
50	0.00072533573049	1.018784322464
100	0.000718695744320	1.009461004069
200	0.000715339856847	1.004747413422
500	0.000713314737917	1.001902985018
1000	0.000712637788046	1.000952159162

Table 2: A numerical illustration of the first omitted coefficient after four terms. Full working-precision output for five truncation orders is included in `artifacts/numerical_checks.csv`.

12 Conclusion

Both formulas in A260306 are proved for every $n \geq 0$. The analytic part identifies the asymptotic coefficients uniquely and proves an $O_R(N^{-R})$ remainder after any fixed number of terms. The algebraic part is finite: a negative-power identity produces the factorial $(3n+2)!$ and the denominator $n+i+1$; the exponential generating function of Stirling numbers gives the double sum; and inclusion–exclusion gives the triple sum.

The main structural answer is the extraction (1.7), or equivalently the residue (4.2). The more elaborate OEIS expressions are exact finite expansions of that simpler object. The recurrence, the associated Stirling-number formula, and the computational checks are consequences and useful cross-checks, not additional assumptions in the proof.

A Minimal exact coefficient generator

The following standalone routine computes all coefficients from c_0 through c_K using (8.5) and (8.6). It uses no symbolic algebra package and performs no floating-point operations.

```
from fractions import Fraction
from math import factorial

def ramanujan_coefficients(K: int) -> list[Fraction]:
    if not isinstance(K, int) or isinstance(K, bool) or K < 0:
        raise ValueError("K must be a nonnegative integer")
    h = [Fraction(1)]
    for m in range(1, 2 * K + 2):
        convolution = sum(
            (h[m-j] * h[j] / (j+1) for j in range(1, m)),
            Fraction(0),
        )
        h.append(
            (h[m-1] / m - convolution) * Fraction(m+1, m+2)
        )
    return [
        Fraction(int(n == 0)) - 2**n * factorial(n) * h[2*n+1]
        for n in range(K + 1)
    ]

for n, coefficient in enumerate(ramanujan_coefficients(10)):
    print(n, coefficient.numerator, coefficient.denominator)
```

The fuller code/verify.py in the archive includes the independent algorithms and validation checks discussed in Section 11. The exact checks were executed with Python 3.13.5; the supplied scripts use syntax supported by Python 3.10 and later.

References

- [1] The OEIS Foundation Inc., *A260306: Numerators in Ramanujan's asymptotic expansion of theta(n)*, entry authored by Vladimir Reshetnikov, November 10, 2015, with subsequent contributions. <https://oeis.org/A260306>. Text record consulted September 20, 2026: <https://oeis.org/search?q=id:A260306&fmt=text>.
- [2] The OEIS Foundation Inc., *A065973: Denominators in an asymptotic expansion of Ramanujan*. <https://oeis.org/A065973>. Consulted September 20, 2026.
- [3] Cormac O'Sullivan, *Ramanujan's approximation to the exponential function and generalizations*, arXiv:2205.08504 [math.NT], 2022. See the introduction and the discussion of associated Stirling numbers in Section 7. <https://arxiv.org/abs/2205.08504>.